

Alvin Yan

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EDUCATION

Boston University – *B.S. in Computer Engineering* | GPA: 3.76

Sept 2023 - May 2027

Relevant Coursework:

- Smart IoT Devices, Embedded Systems, Operating Systems, High-Perf Computing, Cybersecurity

SKILLS

Languages: C/C++, Python, JavaScript | **Embedded:** ESP32, STM32, Raspberry Pi | **Tools:** FreeRTOS, Tensorflow, I2C, UART

WORK EXPERIENCE

BU Terrier Motorsport - Embedded Systems Engineer | Boston, MA

Sept 2024 - Present

- Assisted in developing and documenting an STM32-based CAN-Watchdog to monitor motor control, tractive, and battery systems, reporting system failures, improving reliability through real-time system debugging.
- Reconfigured and validated an OrionBMS-based Accumulator Management System, updating fault thresholds, interlocks, and CAN message mappings to meet Formula SAE Electric safety requirements.
- Implemented USB serial communication between Raspberry Pi and Arduino for real-time sensor data acquisition and logging, ensuring accurate collection under hardware compatibility constraints.

TekFinder - Full Stack Web-Dev Intern | Sydney, Australia

Mar 2025 - May 2025

- Developed an impact visualization software for potential user contributions, increasing platform engagement by ~70%.
- Integrated real-time Google Cloud backend services with interactive frontend components from Figma wireframes, ensuring responsive performance across devices.

TECHNICAL PROJECTS

Autonomous Vehicle Navigation

April 2026 - May 2026

- Trained a CNN regression model on a custom dataset of HSV-segmented camera frames to predict real-time steering angles for autonomous track following on an AgileX Limo.
- Implemented a LiDAR obstacle detection pipeline monitoring a 70-degree frontal arc, triggering emergency stops for dynamic obstacle avoidance within 0.40 meters.
- Built a teleoperation data collection script synchronizing segmented camera frames with steering angle and velocity labels for supervised ML training.

Wearable Activity Tracker & Edge Gateway

March 2026 - April 2026

- Developed C firmware for an ESP32 to acquire and stream high-frequency IMU payloads via UDP to a local edge server.
- Trained a Convolutional Neural Network in TensorFlow.js on a self-collected motion capture dataset and deployed it to a Raspberry Pi for real-time activity classification across standing, walking, and jogging.
- Built and hosted a live web interface on the Raspberry Pi, utilizing Dynamic DNS to expose sensor telemetry and activity predictions to remote clients over the internet.

Sensor Fusion Localization

Feb 2026 - March 2026

- Designed a multi-sensor fusion system on an ESP32 integrating an ADXL343 accelerometer, Garmin LiDAR, and MaxBotix sonar to track 3D positional movement within 5% of actual values.
- Implemented sensor calibration and noise filtering by computing accelerometer offsets at initialization and fusing distance sensor readings to correct double integration drift.
- Built a Node.js visualization server that parsed serial CSV telemetry from the ESP32 and rendered live 3D movement tracking via a p5.js web interface.

Four Legged Walker

Jan 2026 - Feb 2026

- Developed ESP32 firmware using FreeRTOS with separate tasks for servo actuation, alphanumeric I2C display output, and button interrupt handling, coordinated via queues and mutexes.
- Implemented a hardware timer interrupt driven state machine managing transitions between idle countdown and walking modes, driving PWM servo control through a predetermined gait phase sequence.
- Diagnosed and resolved a power delivery bottleneck where four servos exceeded USB current supply limits, redesigning the actuation system to two servos to ensure stable ESP32 operation.